

# Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from 3,900 transactions to guide strategic decisions.



# Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

## Customer Demographics

Age, Gender, Location, Subscription Status

## Purchase & Behavior

Category, Amount, Season, Discount, Frequency, Review Rating, Shipping Type

# Data Preparation in Python

01

---

## Load & Explore

Imported via pandas; used `df.info()` and `.describe()`

02

---

## Clean Missing Data

Imputed Review Rating using median per product category

03

---

## Standardize & Engineer

Renamed to `snake_case`; created `age_group` and `purchase_frequency_days`

04

---

## Load to PostgreSQL

Cleaned DataFrame loaded into database for SQL analysis



# Revenue by Gender & Shipping Type



## Gender Revenue

Compared total revenue generated by male vs. female customers.

## Shipping Comparison

Compared average purchase amounts between Standard and Express shipping.

# Top Products & Discount Users

## Top 5 by Rating

Products with the highest average review ratings.

## Discount-Dependent Products

5 products with the highest percentage of discounted purchases.

## High-Spending Discount Users

Customers who used discounts but spent above average.

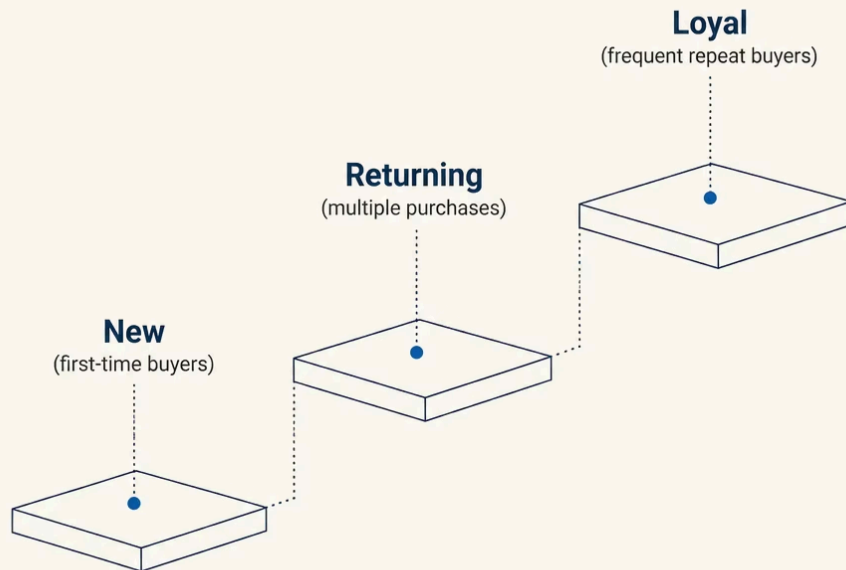


# Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.



# Customer Segmentation



## Segment Classification

Customers classified into **New**, **Returning**, and **Loyal** segments based on purchase history.

## Repeat Buyers & Subscription

Customers with more than 5 purchases are more likely to subscribe.

# Top Products per Category & Revenue by Age



## Top 3 Products per Category

Most purchased products within each category.

## Revenue by Age Group

Total revenue contribution calculated for each age group.

# Power BI Dashboard

Interactive dashboard consolidating all key findings — revenue, segments, product performance, and customer behavior — in one view.



# Business Recommendations



## Boost Subscriptions

Promote exclusive subscriber benefits



## Loyalty Programs

Reward repeat buyers toward "Loyal" segment



## Review Discount Policy

Balance sales boosts with margin control



## Product Positioning

Highlight top-rated, best-selling products



## Targeted Marketing

Focus on high-revenue age groups & express shipping users